

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$2x^2 - 8x - 7 = 0$

One solution to the given equation can be written as $\frac{8-\sqrt{k}}{4}$, where k is a constant. What is the value of k ?

Correct Answer: 120

Rationale

The correct answer is **120**. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$ can be calculated using the quadratic formula and are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -8$, and $c = -7$. It follows that the solutions to the given equation are $x = \frac{8 \pm \sqrt{(-8)^2 - 4(2)(-7)}}{2(2)}$, which is equivalent to $x = \frac{8 \pm \sqrt{64 + 56}}{4}$, or $x = \frac{8 \pm \sqrt{120}}{4}$. It's given that one solution to the equation $2x^2 - 8x - 7 = 0$ can be written as $\frac{8 - \sqrt{k}}{4}$. The solution $\frac{8 - \sqrt{120}}{4}$ is in this form. Therefore, the value of k is **120**.